# Saikeerthi (Rani) Rachavelpula

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Department of Philosophy
Columbia University
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**Areas of Specialty (AOS)**: Metaphysics, Logic
**Areas of Competence (AOC)**: Kant, Epistemology, Ethics, Philosophy of Science

## Education

**Columbia University** *2019–27 (expected)*[*]
Ph.D in Philosophy, GPA: 4.11
**Columbia University** *2015-19*
B.A. in Mathematics with Philosophy Minor, *Magna Cum Laude*
Honors Thesis: *Representing Coxeter Groups* (Advisor: Dusa McDuff)
**University of Cambridge** *2017-18*
Visiting Student, Oxbridge Scholar in Mathematics
**Georgia Institute of Technology** *2014-15*
Calculus III, IV as high school student

**Dissertation**: *Essays on the Antinomies of Space* (abstract on final page)
**Committee**: Achille Varzi (primary), Tamar Lando, Michael Thaddeus (mathematics)

## Publications

[1] "Generating Gunk," *Journal of Philosophical Logic* 55 (2):377–409 (2026).
[2] "Modelling Morality: A Kantian Account of Moral Examples," *Kantian Review* 31 (1):71-92 (2026).

## Presentations

- "On the Difference Between Left and Right Hands," *BLOC Philosophy of Physics Graduate Student Workshop*, *University of Oxford* *2026*
- "Generating Gunk," *CUNY Logic and Metaphysics Workshop* *2025*
- "Introduction to Riemannian Manifolds and Quotient Manifolds," *Brainard Color Vision Lab*, *University of Pennsylvania* *2025*
- "Modelling Morality: A Kantian Account of Moral Examples," *Society for Women in Philosophy (SWIP-NYC)* *2022*
- "Models and Values in Science," *Sorbonne-Columbia Blame and Transition Workshop* *2022*

[*]On official leave of absence for two academic years (2021-2022, 2023-2024).

## Awards & Honors

- Charles Frankel Memorial Fellowship, *GSAS Columbia University* *2024*
- Benjamin Wolstein Fellowship, *GSAS Columbia University* *2022*
- Graduate Essay Prize ($500), *Society for Women in Philosophy (SWIP-NYC)* *2022*
- Provost Diversity Fellowship ($8000), *GSAS Columbia University* *2019*
- James Gutmann Prize in Philosophy ($2000), *Columbia University* *2019*
- Departmental Honors (Top 10% in Mathematics Major), *Columbia University* *2019*
- Oxbridge Scholar (Mathematics), *University of Cambridge* *2017*
- Milstein Scholar, *Columbia University* *2015-19*

## Teaching

*all teaching at Columbia University unless specified otherwise

**Sole Instructor**

Introduction to Symbolic Logic *Summer 2025*

**Teaching Assistant**

Metaphysics (Achille Varzi) *Fall 2026*
Epistemology (Jessica Collins) *Spring 2025*
*Including One Guest Lecture: "Bullshit & Information Content"*
Introduction to Logic (Christopher Prodoehl) *Spring 2023*
Symbolic Logic (Tamar Lando) *Fall 2023 & Fall 2022*
*Including One Guest Lecture: "Formal Proofs & Counterexamples in FOL"*
Ethics (Michele Moody-Adams) *Spring 2021*
Methods & Problems of Philosophical Thought (Akeel Bilgrami) *Fall 2020*

**Grader**

Advanced Introduction to Bioethics (NYU) *Fall 2024 & Fall 2025*
Environmental Ethics (NYU) *Spring 2025*

## Service

- Referee, *Noûs* *2023*
- Colloquium chair, *APA Eastern Division* *2021*
- Co-organizer, *SWIP-NYC (Society for Women in Philosophy in New York)* *2020-23*
- Graduate Student Representative, *Columbia University* *2022-23*
- Co-organizer, *Columbia-NYU Graduate Philosophy Conference* *2019-20*
- Research Assistant (Professor: Katja Vogt), *Columbia University* *2020*
- Book Assistant (Created index, Clarke-Doane's *Morality and Mathematics*), *OUP* *2019*
- Thematic Program in Kähler Geometry, *Notre Dame University* *2018*

- Researcher, Mathematics REU (paper: 'The Category of Mereotopology and Its Ontological Consequences), *The University of Chicago* *2017*

## Outreach

- Differential Geometry for Color Science Working Group, *Brainard Color Vision Lab*, *University of Pennsylvania* *2024-*
- Volunteer Fellow, *Ruttenberg Cancer Treatment Center*, *Mount Sinai* *2025-26*
- Founder/Writer, *Puzzles to Puzzle You (Remastered)* Newsletter *2025*
- Publications Intern, *Artforum International Magazine* *2019*
- Publications Intern, *Museum of Modern Art (MoMA)* *2018*
- Varsity Athlete, NCAA Division I Women's Tennis, *Columbia Unversity* *2015-7*[2]

## Languages

World: French (basic reading), Telugu (native)
Computer: Python, MATLAB, LaTeX

## References

**Achille Varzi**
John Dewey Professor of Philosophy
Columbia University
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**Tamar Lando**
Associate Professor of Philosophy
Columbia University
tal2108@columbia.edu

**Patricia Kitcher**
Roberta and William Campbell Professor Emerita of the Humanities
Columbia University
pk206@columbia.edu

**Jessica Collins**
Associate Professor of Philosophy
Columbia University
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[2]Five-star tennis recruit, ranked top 30 in US. Left team due to injury.

*Essays on the Antinomies of Space* (Dissertation Abstract)

Metaphysical theories of space provide the conceptual foundation for formal theories of spatial representation and reasoning, underpinning modern approaches to computer vision, spatial cognition, and robotics. Since antiquity, however, philosophers have observed that the nature of space itself gives rise to deep and seemingly irresolvable paradoxes. Rather than viewing these paradoxes as mere historical curiosities or obstacles to scientific progress, my dissertation argues that they are indispensable guides to more adequate theories of space. Drawing on tools from geometric analysis, I construct philosophically novel and mathematically rigorous resolutions to three classic spatial paradoxes: (1) Anaxagoras's problem of gunk, (2) Zeno's paradox of measure, and (3) Kant's worries about incongruent counterparts.

The first chapter investigates an apparent conflict between two intuitions about spatial structure: that spatial objects are *gunky* (i.e. infinitely divisible) and that they are *well-founded* (i.e. ultimately grounded in fundamental entities). Utilizing a result from Fourier analysis called Weyl's Equidistribution Theorem, I show that these intuitions are compatible by constructing a model in which every entity has proper parts but the entire structure is generated by a single fundamental entity. I further show that this result revives an underappreciated conception of metaphysical fundamentality inspired by Nelson Goodman's work on generation.

The second chapter addresses a problem from Aristotle's *On Generation and Corruption*: How can infinitely many unextended points compose an extended continuum? This problem is sometimes called Zeno's paradox of measure, and its standard solution invokes mathematical measure theory. The standard solution, however, has been criticized as merely formal and not generalizable to higher-dimensional cases. Drawing on dimension theory, I ground measure-theoretic accounts of spatial extension in a more fundamental notion of spatial occupation, formalized in terms of Hausdorff dimension. This framework not only resolves Aristotle's original paradox and its higher-dimensional manifestations; it also sheds light on contemporary problems of spatial measurement, including the so-called coastline paradox.

The final chapter turns to Kant, whose antinomies of pure reason inspired both the dissertation's title and its general spirit. Whereas Kant's second antinomy concerns the problem of gunk, this chapter addresses Kant's writings on incongruent counterparts, or handed objects such as left and right hands. In particular, I examine his 1768 argument that handed objects establish the independent reality of space over and above the material objects within it. Against recent attempts to ground Kant's argument in a topological notion of handedness, I develop a projection-theoretic account that distinguishes handed objects without presupposing the independent existence of space. This framework is motivated by my work in pure mathematics, where I introduce the first anisotropic variant of Buffon's Needle Problem. I prove several mathematical results about this new problem which reveal unexpected connections between projection theory, anisotropic geometry, and the metaphysics of spatial orientation.

Taken together, these results show that paradoxes or 'pathological' cases are indispensable tools for philosophical inquiry. In mathematics, the limits of existing theories are often exposed by 'pathologies' such as Peano's space-filling curve or Weierstrass's everywhere-continuous, nowhere-differentiable function. Such cases push mathematicians to develop better theories. My dissertation shows that philosophical paradoxes play a similar role, giving us new ways of understanding familiar spatial phenomena, with implications to broader questions about the relationship between formal representation and metaphysical reality.

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